

# Adaptive feed-forward compensation for hybrid control with acceleration time waveform replication on electro-hydraulic shaking table



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## ABSTRACT

This article presents a high fidelity acceleration time waveform replication (TWR) on an electro-hydraulic shaking table (EHST) using the hybrid control combined with an offline feed-forward compensator and an online adaptive inverse control (AIC). This study applies the acceleration and velocity feedback to improve the steady performance of the EHST, and employs the system inverse transfer function (ITF) of the acceleration closed-loop system to extend the frequency bandwidth and a modeling error compensator to improve the dynamic characteristics. Moreover, the investigation utilizes a Zero Phase Error Tracking Controller to improve the accuracy of the designed ITF. The proposed hybrid controller also utilizes an online AIC for a high fidelity TWR after that the dynamic characteristics has been improved by the feedback and feed-forward controllers. Thus, the proposed hybrid control strategy combines the merits of the offline feed-forward compensation and online AIC. Performance analysis and comparison of the experimental results demonstrate that better replication accuracy with the proposed hybrid control can be achieved in experiments on an actual EHST.

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## 1. Introduction

Electro-hydraulic shaking tables (EHST) are extensively employed in several industries to real-time replicate actual situations in the civil and architectural engineering (Ji, Kajiwar, & Nagae, 2009), the automotive industry (Kim, Xuan, & Kim, 2008; Plummer, 2007) and the earthquake resistance testing (Severn, 2011), etc. The time waveform replication (TWR) is used in practice to determine whether a test article can remain operational and retain its structure integrity after being subjected to a specific shock and vibration environment (Underwood, Tu, & San, 2001). However, the acceleration output response of the EHST system would not be intended in its magnitude and phase if there is not any compensation for the dynamic characteristics, because it may be affected by several factors such as physical system parameters and the dynamic characteristics of the payload (Thomas & Donald, 1997).

In order to extend the frequency bandwidth and to improve the stability of the EHST control system, Three Variable Controller

(TVC) (Tagawa & Kajiwar, 2007; Xu, Hua, & Han, 2008) is widely employed in the EHST acceleration closed-loop system as a basic servo controller. The idea is to replace the unwanted EHST dynamics with the designer's own desired dynamics (Zhang, Alleyne, & Zheng, 2005). As a benchmark controller for many large shaking tables (Gizatullin & Edge, 2006; Stoten & Shimizu, 2007; Tagawa & Kajiwar, 2007), the TVC has drawn a lot of attention. In order to compensate the nonlinearities associated with pneumatic actuator systems, Jihong Wang, Pu, and Moore (1999) employed a TVC feedback controller to improve the velocity tracking accuracy. Shen, Zheng, and Ye (2011) proposed an equivalent principle between the TVC feed-forward controller and the position closed-loop inverse model.

However, the frequency bandwidth of the acceleration closed-loop is limited by the strongly nonlinear behavior of the EHST system (Garagic & Srinivasan, 2004) and affected by the natural angular frequency and damping ratio. Various compensation control strategies have been proposed to obtain a high fidelity TWR. Maddaloni, Ryu, and Reinhorn (2011) proposed the feed-forward ITF compensation method as a key technique which was employed to obtain better waveform replication results. Cheng and James (2009a, 2009b) developed an inverse compensation to minimize the effect of the actuator delay in a real-time hybrid

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simulation. Lee, Park, and Min (2007) employed an acceleration ITF of the shaking table to cancel out the dynamic characteristics. Besides, a cascade of compensators was designed to attenuate the shaker structural resonances and to obtain an asymptotic reference tracking (Della & Gründling, 2008). However, the identified acceleration closed-loop model may be a Non-Minimum Phase system caused by sensors and actuators (Rigney, Lucy, & Lawrence, 2010), and its inverse model is an unstable system. To overcome this problem, the adaptive Finite Impulse Response was employed in practical applications (Sharq, 2005; Shen, Lv, Cong, & Han, 2011). For example, Rigney et al. (2010) relied on a Recursive Least Square parameter estimation algorithm to identify the closed-loop dynamics, and then used the zero-order Taylor series approximate inverse technique to synthesize its inverse model. Shen, Lv, Cong, and Han (2012) proposed an improved internal model control and a 2 degree-of-freedom (DOF) real-time feed-back controller to reduce the modeling error.

In order to improve the accuracy of the acceleration TWR, MTS Systems Corporation developed a commercial package named

Remote Parameter Control in 1977, which employed the ITF matrix of the controlled plant and was widely used in automotive industries. In fact, the iterative control through repetitive excitations has become an essential tool in the structural testing and seismic testing fields (Cuyper & Verhaegen, 2002; Cuyper, Verhaegen, & Swevers, 2003; Goele & Jan, 2010; Plummer, 2007; Tagawa & Kajiwara, 2007).

These feed-forward compensation methods present some specific advantages: (a) the effect is restricted to the input channel to which it is added (Goele & Jan, 2010), and (b) the system stability will not be affected, and disadvantages: (a) repetitive excitations may break the specimen before it is subjected to the desired excitation level (Goele & Jan, 2010), (b) the system uncertainty and the modeling error between the estimated model and actual plant will influence the TWR accuracy and (c) the estimated ITF may be mismatched with the actual plant during real-time testing when some factors influence it.

Thus, some investigators expanded from the offline feed-forward ITF compensation to the online adaptive control, which have recently attracted attention because the payload dynamics may be variable and it may change the frequency response characteristics of the EHST system. These approaches include robust control (Uchiyama, Mukai, & Fujita, 2009), minimal control synthesis (Gizatullin & Edge, 2006; Neild, Drury, & Stoten, 2005; Stoten & Gómez, 2001; Stoten & Neild, 2003; Stoten & Shimizu, 2007), the adaptive inverse control (AIC) (Karshenas, Dunnigan & Williams, 2000; Salehzadeh-Nobari, Chambers, & Green, 2000), the amplitude and phase control and the self-tuning control (Ghalibafian, Bhuyan, & Ventura, 2004; Plummer, 2007). However, these adaptive algorithms have their own advantages and disadvantages, e.g. high quality waveform replication accuracy can be obtained after it converge to the optimal solution, but it is possible to exhibit poor transient response in initiating, especially when the frequency range of the desired acceleration signal exceeds the frequency bandwidth of the acceleration closed-loop system.

The feed-forward inverse compensation and adaptive control are popular on the EHST and some hybrid and combined control strategies were used to improve the acceleration TWR accuracy. Cuyper and Verhaegen (2002), Cuyper et al. (2003), Vaes, Swevers,



Fig. 1. Real-time experimental system of the EHST.

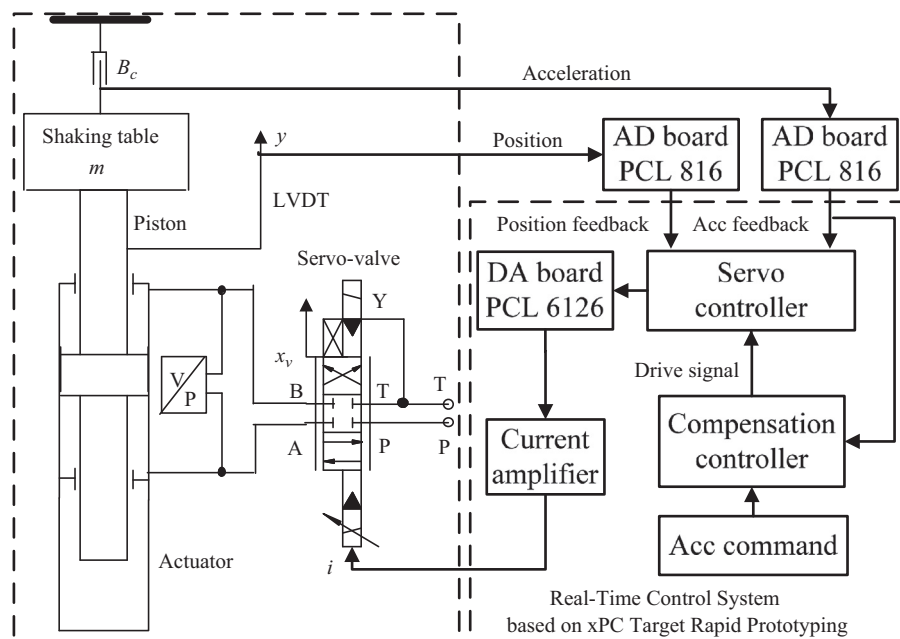


Fig. 2. Block schematic of the experimental EHST system.



performance are given in Section 4, which demonstrate the effectiveness of the proposed hybrid control strategy. Section 5 concludes the main points and contributions.

## 2. Experimental system set-up

The proposed control strategy is implemented in the EHST real-time experimental system shown in Fig. 1. The EHST system consists of a moving steel platform, eight servo-controlled hydraulic actuators and eight servo-valves, a real-time computer controller, and a hydraulic power supply system. The double ended hydraulic actuator has an 80 mm bore and a 60 mm rod and the servo-valve is a two stage valve with a 38 L/min flow capacity at 7 MPa supply pressure manufactured by Moog, Inc. (type: G761-3004). The servo-valve is driven by the input current supplied by the servo controller in Fig. 2. The technical specifications of the EHST system are listed in Table 1. Each hydraulic actuator is attached a Linear Variable Differential Transformer (LVDT) to measure the actuator's displacement and the eight channels of acceleration output response are measured with eight accelerometers mounted on the drive rod. The mathematical model of the physical system can be obtained from the references (Plummer, 2008; Shen, Lv et al., 2011). The data acquisition and implementation of the digital controller are conducted on xPC target based on MATLAB/Simulink rapid prototyping (Huyanan & Sims, 2007). Data acquisition is accomplished by Advantech A/D card PCL-816 and D/A card PCL-6126. The sampling interval of the digital controller is selected as 0.001 s. The procedure of the proposed control strategy is programmed in MATLAB/Simulink, and then is compiled to a real-time PC with Microsoft Visual Studio.NET for implementation.

## 3. Hybrid controller and its design

The proposed hybrid controller framework is shown in Fig. 3. The hybrid controller consists of a feedback controller, a feed-forward compensator, a modeling error compensator and an online AIC. For the feedback controller, a TVC feedback controller is used to improve the stable performance of the EHST. For the feed-forward compensator, an offline designed ITF and a modeling error compensator are utilized to extend the acceleration frequency bandwidth and to reduce the modeling error between the estimated model and actual system, respectively. An online AIC is added sequentially to further improve the TWR performance.

So, the proposed control strategy combines the merits of the offline feed-forward compensator and online AIC. A manual switch is utilized to switch into the offline feed-forward compensator case or proposed hybrid controller case during tests according to different phases and test requirements.

The proposed hybrid control strategy has a simple structure and its actual implementation can be readily realized by means of a MATLAB/Simulink model and its xPC target.

### 3.1. Design of the TVC

The TVC is a commonly used controller for shaking tables (Tagawa & Kajiwar, 2007). A block diagram of the TVC is given in Fig. 4. TVC is a 2 DOF controller including a feedback controller and a feed-forward controller. TVC feedback states, including displacement feedback signal  $y_d$  measured by the LVDT, acceleration feedback signal  $y_a$  measured by accelerometer mounted on the table, and velocity feedback signal  $y_v$  synthesized from the measured displacement processed by a low-pass-filter and a differential, and the measured acceleration processed by a high-pass-filter and an integral (Tagawa & Kajiwar, 2007), are used to improve the stable performance of the EHST. TVC feed-forward states, including  $R_d$ ,  $R_v$  and  $R_a$  are generated by the reference signal generator to extend the acceleration frequency bandwidth and to improve the TWR performance by tuning three feed-forward parameters ( $K_{dr}$ ,  $K_{vr}$  and  $K_{ar}$ ).

Referring to Fig. 4, the acceleration closed-loop transfer function  $G_a(s)$  can be expressed as

$$G_a(s) = \frac{K_u(K_{dr} + K_{vr}s + K_{ar}s^2)}{d_0(s^2/\omega_0^2 + (2\zeta_0/\omega_0)s + 1)} \frac{1}{(s/\omega_r + 1)(s^2/\omega_{nc}^2 + (2\xi_{nc}/\omega_{nc})s + 1)} \quad (1)$$

where  $\omega_r$  is the desired frequency bandwidth of the acceleration response,  $\omega_{nc}$  is defined as  $\omega_{nc} = 1.2\omega_h$ ,  $\omega_h$  is the natural angular frequency of the hydraulic actuator, and  $\xi_{nc}$  is defined as  $\xi_{nc} = 0.7$ ,  $K_u$  is the acceleration gain,  $\zeta_0$  is the damping ratio of the acceleration control,  $\omega_0$  is the original frequency of acceleration, and  $d_0$  is defined as  $d_0 = \omega_0^2$ .

The TVC feed-forward controller is used to cancel the system pole at  $\omega_{nc}$ . Thus, the following equation can be obtained (Shen et al., 2012):

$$F(s) = K_{dr} + K_{vr}s + K_{ar}s^2 = \frac{s^2}{\omega_{nc}^2} + \frac{2\xi_{nc}}{\omega_{nc}}s + 1 \quad (2)$$

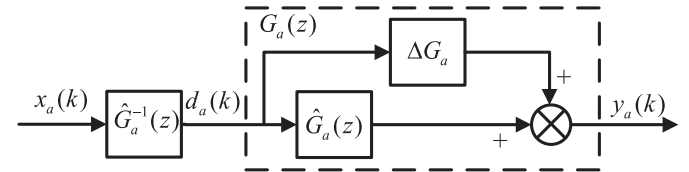


Fig. 5. Acceleration feed-forward compensation with model error taken into consideration.

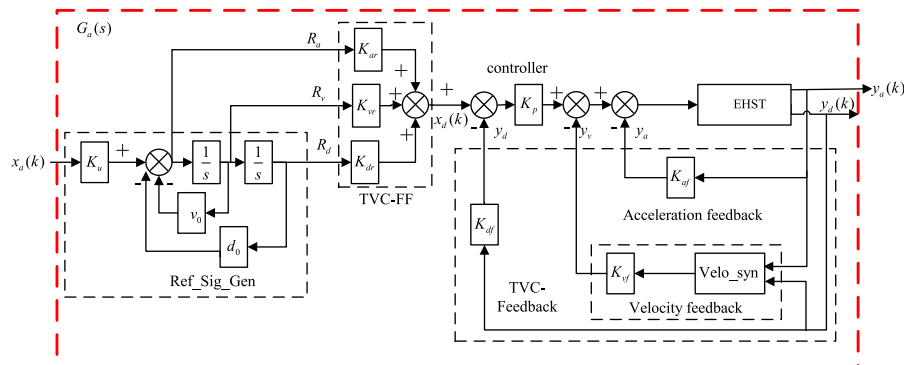


Fig. 4. Block diagram representation of the TVC.

Eq. (1) can be rewritten as

$$G_a(s) = \frac{K_u}{d_0(s^2/\omega_0^2 + (2\zeta_0/\omega_0)s + 1)(s/\omega_r + 1)} \quad (3)$$

### 3.2. Design of the feed-forward ITF

The acceleration output response  $y_a(k)$  may not perfectly follow the acceleration command  $x_a(k)$  due to the strongly nonlinear behavior and the narrow frequency bandwidth of the EHST system. Thus, a feed-forward ITF compensator of the acceleration closed-loop is employed to improve the accuracy of the TWR. The block diagram of the ITF compensator is given in Fig. 5, where the TVC control strategy is retained as an inner-loop component.

The discrete ITF  $\hat{G}_a^{-1}(z)$  in the figure is estimated by the Recursive Extended Least Square algorithm and is further offline designed by a Zero Phase Error Tracking Controller, which determines the required closed-loop input  $d_a(k)$ . The acceleration feed-forward inverse model  $\hat{G}_a^{-1}(z)$  converts the acceleration reference signal  $x_a(k)$  into an appropriate acceleration drive signal  $d_a(k)$  to make the acceleration TWR error smaller. Thus the transfer function from  $x_a(k)$  to  $y_a(k)$  can be further derived as

$$G_a(s) = \frac{\hat{G}_a^{-1}(s)K_u}{d_0((s^2/\omega_0^2) + (2\zeta_0/\omega_0)s + 1)((s/\omega_r) + 1)} \quad (4)$$

where  $\hat{G}_a^{-1}(s)$  is the frequency domain form of  $\hat{G}_a^{-1}(z)$ .

The estimated system model  $\hat{G}_a(z)$  differs from the actual plant  $G_a(z)$  and the difference can be described by a modeling error  $\Delta G_a$  as represented in Fig. 5. This  $\Delta G_a$  may include parameter uncertainties in  $G_a(z)$  and the modeling error between  $\hat{G}_a(z)$  and  $G_a(z)$ . The modifier  $\wedge$  on both  $\hat{G}_a^{-1}(z)$  and  $\hat{G}_a(z)$  indicates that these quantities are estimated and not the exact inverse and model of  $G(z)$ , respectively.

The following relationship can be derived from Fig. 5:

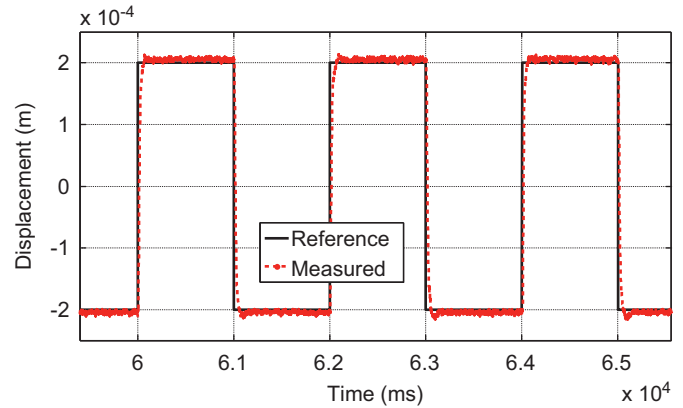
$$Y_a(z) = \hat{G}_a^{-1}(z)(\hat{G}_a(z) + \Delta G_a)X_a(z) = X_a(z) + \hat{G}_a^{-1}(z)\Delta G_a X_a(z) \quad (5)$$

where  $Y_a(z) = Z[y_a(k)]$  and  $X_a(z) = Z[x_a(k)]$ .

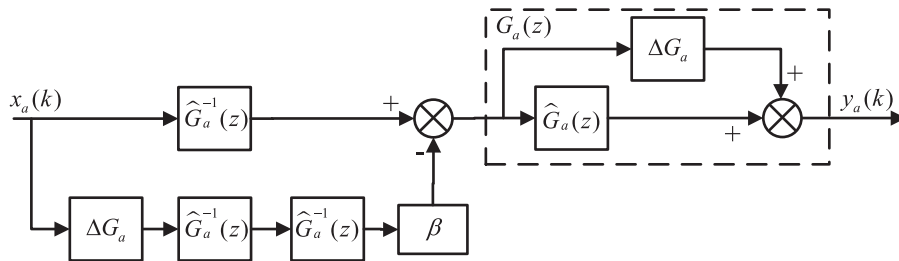
**Table 2**

Experimental parameters.

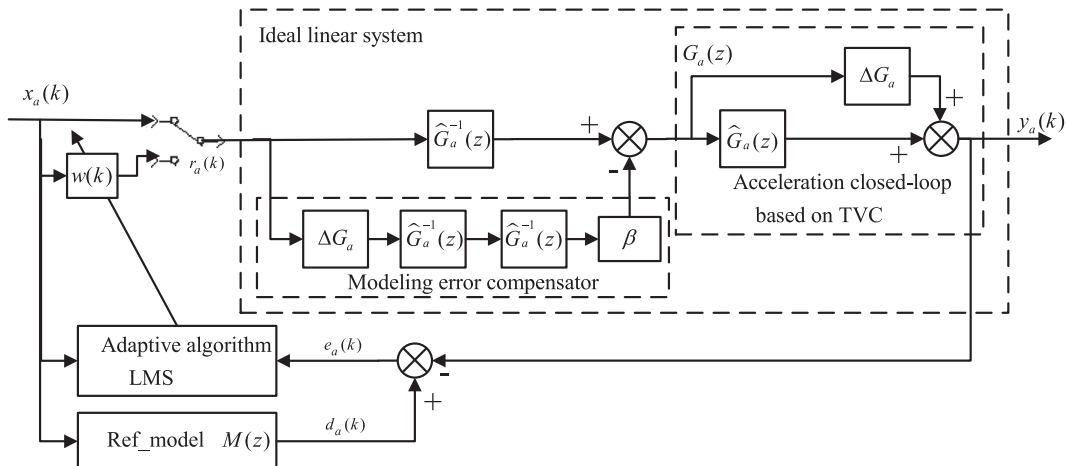
| Parameters | Value | Parameters | Value  | Parameters | Value        |
|------------|-------|------------|--------|------------|--------------|
| $K_{af}$   | 0.02  | $K_{ar}$   | 0.0016 | $K_p$      | 16.5         |
| $K_{vf}$   | 0.1   | $K_{vr}$   | 2.9    | $\omega_0$ | 3.14 (rad/s) |
| $K_{df}$   | 133.3 | $K_{dr}$   | 133.3  | $\zeta_0$  | 0.6          |



**Fig. 8.** Position output response with square wave based on TVC feedback controller.



**Fig. 6.** Block diagram representation of the proposed model error compensator.



**Fig. 7.** Block diagram representation of the proposed hybrid control.

As can be seen in the above formula,  $\Delta G_a$  and  $\hat{G}_a^{-1}(z)$  are important factors that influences the TWR accuracy and the acceleration TWR error can be presented as  $\hat{G}_a^{-1}(z)\Delta G_a x_a(k)$ .

In order to improve the TWR accuracy, a modeling error compensator is proposed. The block diagram is given in Fig. 6. Thus, the acceleration output response  $Y_a(z)$  can be expressed as

$$\begin{aligned} Y_a(z) &= [\hat{G}_a^{-1}(z) - \beta \Delta G_a \hat{G}_a^{-1}(z) \hat{G}_a^{-1}(z)] (\hat{G}_a(z) + \Delta G_a) X_a(z) \\ &= X_a(z) + \hat{G}_a^{-1}(z) \Delta G_a X_a(z) - \beta \Delta G_a \hat{G}_a^{-1}(z) X_a(z) \\ &\quad - \beta (\hat{G}_a^{-1}(z) \Delta G_a)^2 X_a(z) \end{aligned} \quad (6)$$

with the relationship  $0 \leq \beta \leq 1$ .

The acceleration TWR error  $E_a(z)$  can be written as the following:

$$\begin{aligned} E_a(z) &= X_a(z) - Y_a(z) = \beta \Delta G_a \hat{G}_a^{-1}(z) X_a(z) + \beta (\hat{G}_a^{-1}(z) \Delta G_a)^2 X_a(z) \\ &\quad - \hat{G}_a^{-1}(z) \Delta G_a X_a(z) \end{aligned} \quad (7)$$

Within the desired frequency range, we have the following relationship:

$$|\hat{G}_a^{-1}(z) \Delta G_a|^2 \leq |\hat{G}_a^{-1}(z) \Delta G_a| \leq 1 \quad (8)$$

Based on Eqs. (7) and (8), the following relationship can be derived:

$$\begin{aligned} E_a(z) - \hat{G}_a^{-1}(z) \Delta G_a X_a(z) &= (\beta - 1) \Delta G_a \hat{G}_a^{-1}(z) X_a(z) \\ &\quad + \beta (\hat{G}_a^{-1}(z) \Delta G_a)^2 X_a(z) - \hat{G}_a^{-1}(z) \Delta G_a X_a(z) \\ &\leq (\beta - 1) \Delta G_a \hat{G}_a^{-1}(z) X_a(z) + (\beta - 1) (\hat{G}_a^{-1}(z) \Delta G_a)^2 X_a(z) \\ &\leq 0 \end{aligned}$$

thus, the acceleration TWR error  $E_a(z)$  will not be larger than  $\hat{G}_a^{-1}(z) \Delta G_a X_a(z)$ . Therefore, the proposed compensator improves the acceleration TWR accuracy.

### 3.3. System identification and inverse model design

The accuracy of the TWR will be affected if the identified inverse model is a plant-model mismatch (Mustafa, 2002). Referring to Appendices A and B,  $\hat{G}_a(z)$  and  $G_a^{-1}(z)$  can be obtained based on the Recursive Extended Least Square algorithm and the Zero Phase Error Tracking control as following:

$$\hat{G}_a(z) = \frac{z^{-d} \hat{B}_a(z) \hat{B}_u(z)}{\hat{A}(z)} \quad (9)$$

$$G_a^{-1}(z) = \alpha \frac{\hat{A}(z) \hat{B}_u(z^{-1})}{z^L \hat{B}_a(z)} G_c(z) \quad (10)$$

$$G_c(s) = e^{LTs} \cong \frac{1 + L \times T \times s/1 + (L \times T \times s)^2/2 + (L \times T \times s)^3/6 + (L \times T \times s)^4/24}{((s/1000) + 1)^4} \quad (11)$$

where  $d$  is the number of pure time step delays,  $\hat{B}_a(z)$  is the numerator polynomial including all plant Minimum Phase zeros,  $\hat{B}_u(z)$  contains all plant Non-Minimum Phase zeros,  $G_c(z)$  is the  $z$ -transform of the compensator  $G_c(s)$ .

### 3.4. Design of the adaptive feed-forward compensator

Referring to Fig. 7 and Appendix C, the optimum solution of the proposed control scheme can be given as

$$w(k)_{opt} = (1 + \hat{G}_a^{-1}(z) \Delta G_a - \beta \hat{G}_a^{-1}(z) \Delta G_a - \beta (\hat{G}_a^{-1}(z) \Delta G_a)^2)^{-1} z^{-L} \quad (12)$$

The weight update of the improved Normalized Least Mean Square algorithm is given as (Vaes et al., 2007)

$$w(k+1) = w(k) + \rho e_a(k) x_a(k) \quad (13)$$

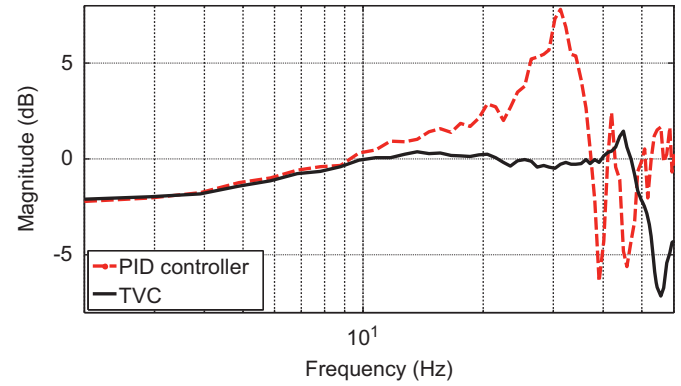


Fig. 10. Frequency response characteristics of acceleration TWR with PID and TVC controllers.

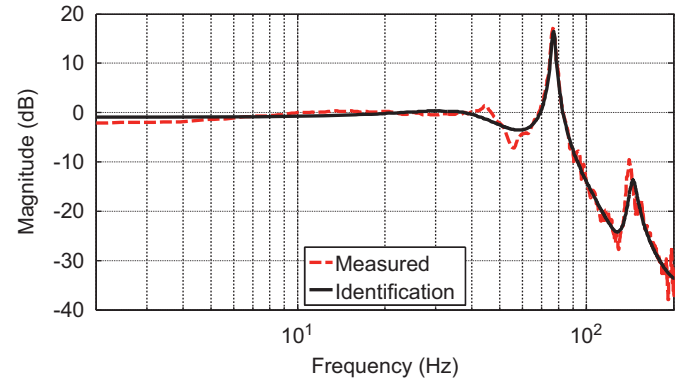


Fig. 11. Experimental and identified acceleration model magnitude frequency characteristics.

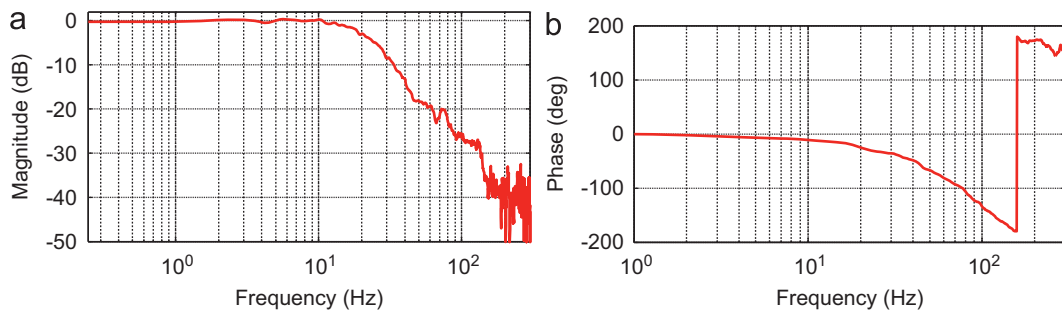


Fig. 9. Experimental position frequency property based on TVC feedback. (a) Position magnitude property and (b) position phase property.

where  $\rho = \mu / (\alpha / (\lambda + \sum_{i=k-l+1}^k e_a^2(k)) + (1-\alpha)\|x_a(k)\|^2)$ ,  $\alpha$  ( $0 < \alpha < 1$ ) and  $\lambda$  ( $0 < \lambda < 1$ ) are small positive numbers,  $l$  ( $0 < l < k$ ) is the length of data window, and  $\mu$  is the step-size parameter.  $\alpha$  is used to fine-tune the step-size  $\mu$  and  $\lambda$  is employed to prevent the denominator equal to zero.

The optimal solution and stability condition for step  $\mu$  have been deduced. However, the convergence performance of the proposed hybrid controller will be directly related with the phase bandwidth of the acceleration closed-loop. Generally assume that the acceleration output response delay is the sampling period integer times, and this delay has been known. The real transfer function error can be obtained according to the known delay, which is an important condition for evaluating the stability of the AIC algorithm. So, the conclusion is that the phase frequency bandwidth of the controlled plant can not be greater than  $90^\circ$  at the frequency of interest. If the frequency range of acceleration command signal is greater than the phase bandwidth ( $-90^\circ$  responding to frequency) of the controlled plant, the Least Mean Square algorithm will search in the wrong direction and finally diverges (Widrow, McCool, & Larimore, 1976). The convergence rate and divergence condition of the Least Mean Square algorithm were given in the reference (Aarón, 2005; Dennis, 1980; Michael, William, & William, 2003). From Fig. 7, the convergence condition of the proposed hybrid controller can be represented as

$$\begin{cases} \angle(1 + \hat{G}_a^{-1}(z)\Delta G_a) < 90^\circ, & \text{if } \beta = 0 \\ \angle(1 - (\hat{G}_a^{-1}(z)\Delta G_a)^2) < 90^\circ, & \text{if } \beta = 1 \end{cases} \quad (14)$$

As can be seen in the above formula, as long as (14) is satisfied, the proposed hybrid control strategy will converge to the optimal solution. However, the stability and convergence rate

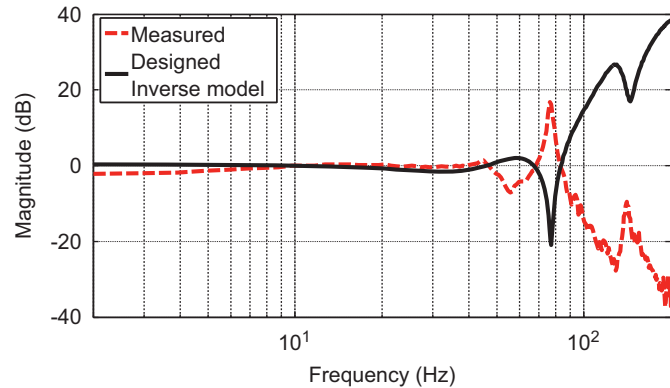


Fig. 12. Experimental and designed acceleration inverse model magnitude frequency characteristics.

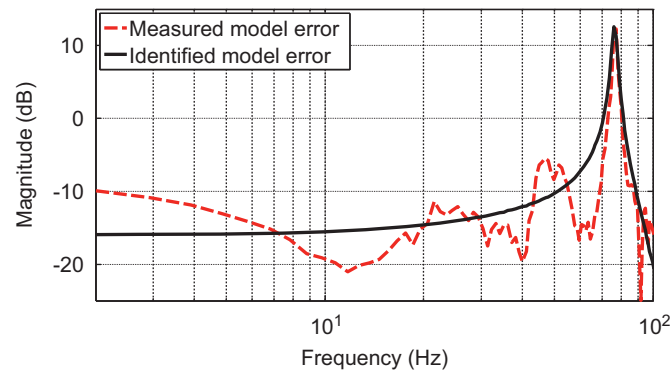


Fig. 13. Experimental and identified  $\Delta G_a$  magnitude frequency characteristics.

will be affected by the offline designed acceleration ITF  $\hat{G}_a^{-1}(z)$  and its modeling error  $\Delta G_a$ .

## 4. Experimental results and analysis

### 4.1. Servo control system experimental results

A series of experimental tests have been conducted to evaluate the performance of the proposed approach. The servo control system of EHST based on conventional PID and TVC is tuned in the first phase. A TVC feedback controller (by tuning gains  $K_{af}$ ,  $K_{vf}$  and  $K_p$ ) is used to adjust the stability of the EHST system using position square wave response based on that the overshoot and setting times are reduced as quickly as possible. The actual tuned experimental gains of the TVC feedback controller are listed in Table 2. Fig. 8 shows the position square wave response, which displays good TWR performance. Fig. 9 shows the position frequency response between the position drive signal  $x_d(k)$  and position output response  $y_d(k)$ . It can be seen from the figure that the frequency bandwidth is approximately 15 Hz within  $-3$  dB.

In order to verify the effectiveness of the TVC in the EHST, acceleration TWR with PID and TVC tests are carried out. Fig. 10 shows the acceleration closed-loop system frequency response between the acceleration command  $x_a(k)$  and acceleration output response  $y_a(k)$ . In the figure, the dashed line shows the experimental results of the conventional PID and reference signal generator (with the parameters  $K_p=16.5$ ,  $K_{af}=K_{vf}=K_{ar}=K_{vr}=0$ ,  $K_{df}=K_{dr}=133.3$ ) and the continuous line shows the experimental results of the TVC with both feedback and feed-forward controller (with the gain values shown in Table 2). It can be seen from the figure that the frequency bandwidth of the acceleration closed-loop system improves from approximately 15 Hz with the magnitude within  $\pm 2$  dB based on the conventional PID controller to approximately 50 Hz with the magnitude within the range of only  $-3$  dB to 2 dB based on the TVC.

Consequently, the acceleration TWR performance is poor with PID and a satisfactory experimental result (a frequency bandwidth of 50 Hz) is obtained by adding the TVC feedback controller and TVC feed-forward controller. The experimental results demonstrate the potential of the TVC to achieve better performance.

### 4.2. Experimental results of feed-forward compensation

In the tests of the second phase, the feed-forward ITF and its modeling error compensator are employed to improve the accuracy of TWR on the EHST. Firstly, the acceleration closed-loop system identification and its inverse model need to be identified and designed for the feed-forward compensation test. An identification test is carried out with a 2–200 Hz random acceleration

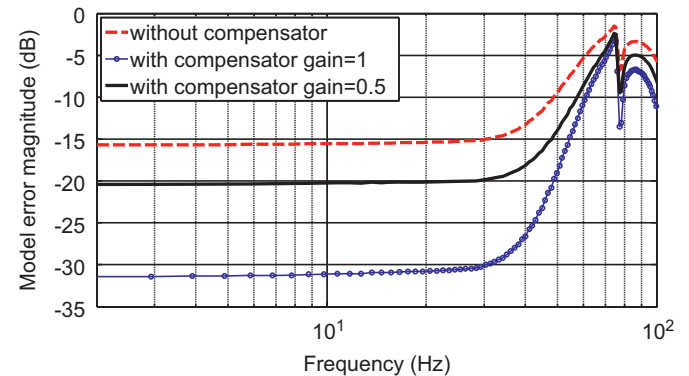


Fig. 14. Ideal tracking error magnitude property with different  $\beta$ .

reference signal with all parameters shown in Table 2. The acceleration reference signal data and measured experimental data are acquired to estimate the model using Recursive Extended Least Square algorithm and further to design their inverse model, respectively. A Controlled Auto-Regressive Moving Average model with 9 orders is selected as the acceleration closed-loop system model in order to obtain an estimated model of high accuracy. The identified acceleration model can be represented as

$$\hat{G}_a(z) = \frac{N}{D} \quad (15)$$

where

$$\begin{aligned} N &= -0.008559z^9 + 0.07688z^8 - 0.2619z^7 + 0.6274z^6 - 1.0413z^5 \\ &\quad + 1.224z^4 - 1.01813z^3 + 0.5954z^2 - 0.2299z + 0.04822 \\ D &= z^9 - 4.1051z^8 + 7.345z^7 - 7.4067z^6 + 5.2519z^5 - 4.4613z^4 \\ &\quad + 4.949z^3 - 4.028z^2 + 1.7942z - 0.3258 \end{aligned}$$

Fig. 11 presents the comparison of experimental and estimated acceleration transfer function magnitude frequency characteristics. The figure is calculated from the reference input or estimated model output response using the 'tfestimate' command in

MATLAB. The measured signals are smoothed using a Hanning window in order to minimize the problem of leakage. It can be observed from Fig. 11 that the estimated model is well approximated by the Recursive Extended Least Square algorithm.

The zeros and poles are obtained as following in order to verify whether the estimated model is Non-Minimum Phase system or not:

$$\hat{G}_a(z) = \frac{N'}{D'} \quad (16)$$

where

$$\begin{aligned} N' &= -0.0085594(z^2 - 1.609z + 0.7607)(z^2 - 1.257z + 0.8616)(z \\ &\quad - 5.032)(z^2 - 0.4273z + 0.5441)(z^2 - 0.6574z + 3.14) \\ D' &= (z - 0.5679)(z^2 - 1.726z + 0.7889)(z^2 - 1.763z + 0.9916)(z^2 \\ &\quad + 1.149z + 0.7745)(z^2 - 1.198z + 0.9467). \end{aligned}$$

It can be seen that the estimated acceleration closed-loop system is a Non-Minimum Phase system (there are three zeros  $z=5.032$  and  $z=0.329 \pm 1.74i$  outside the unit circle) and its inverse is an unstable system.

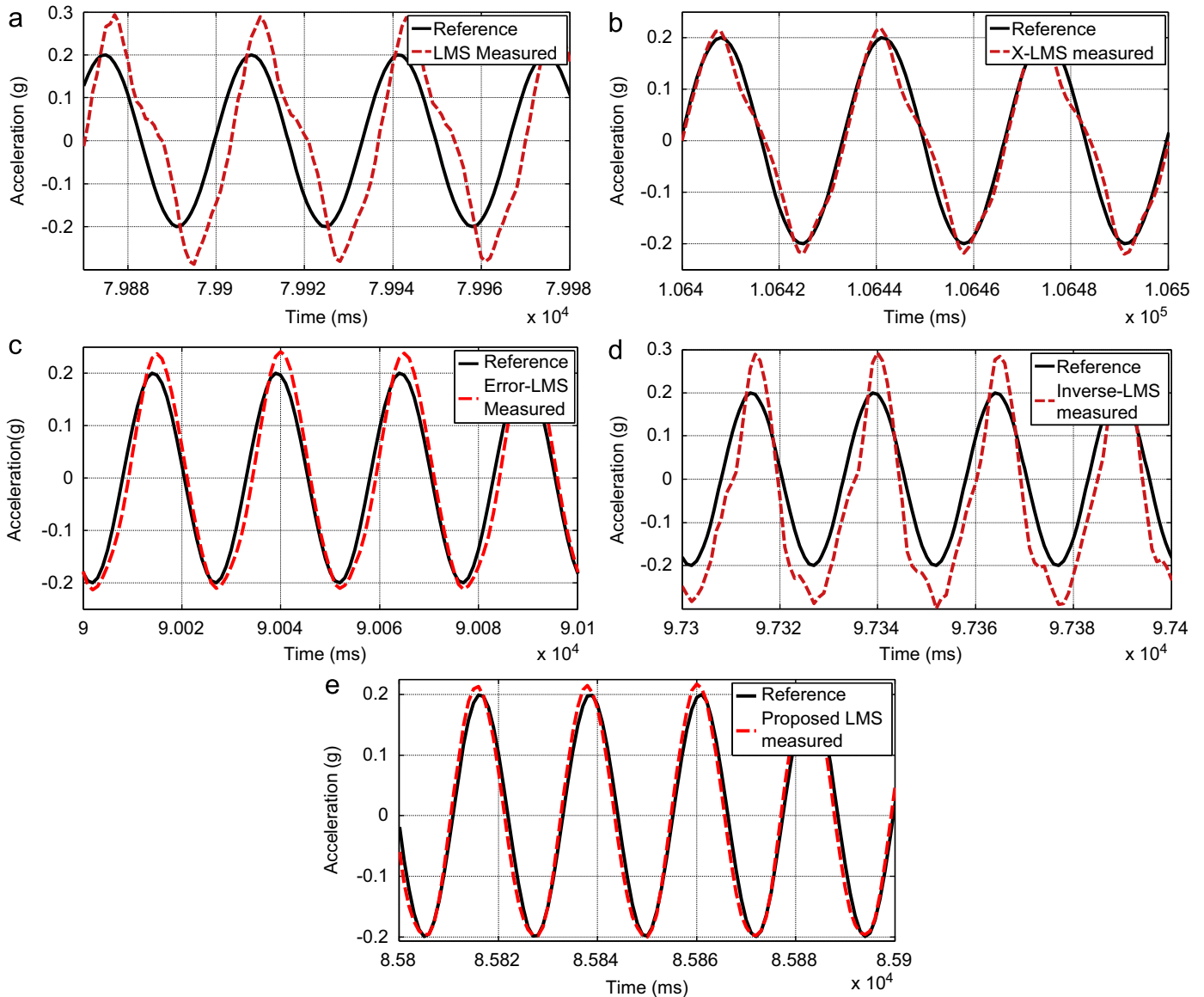
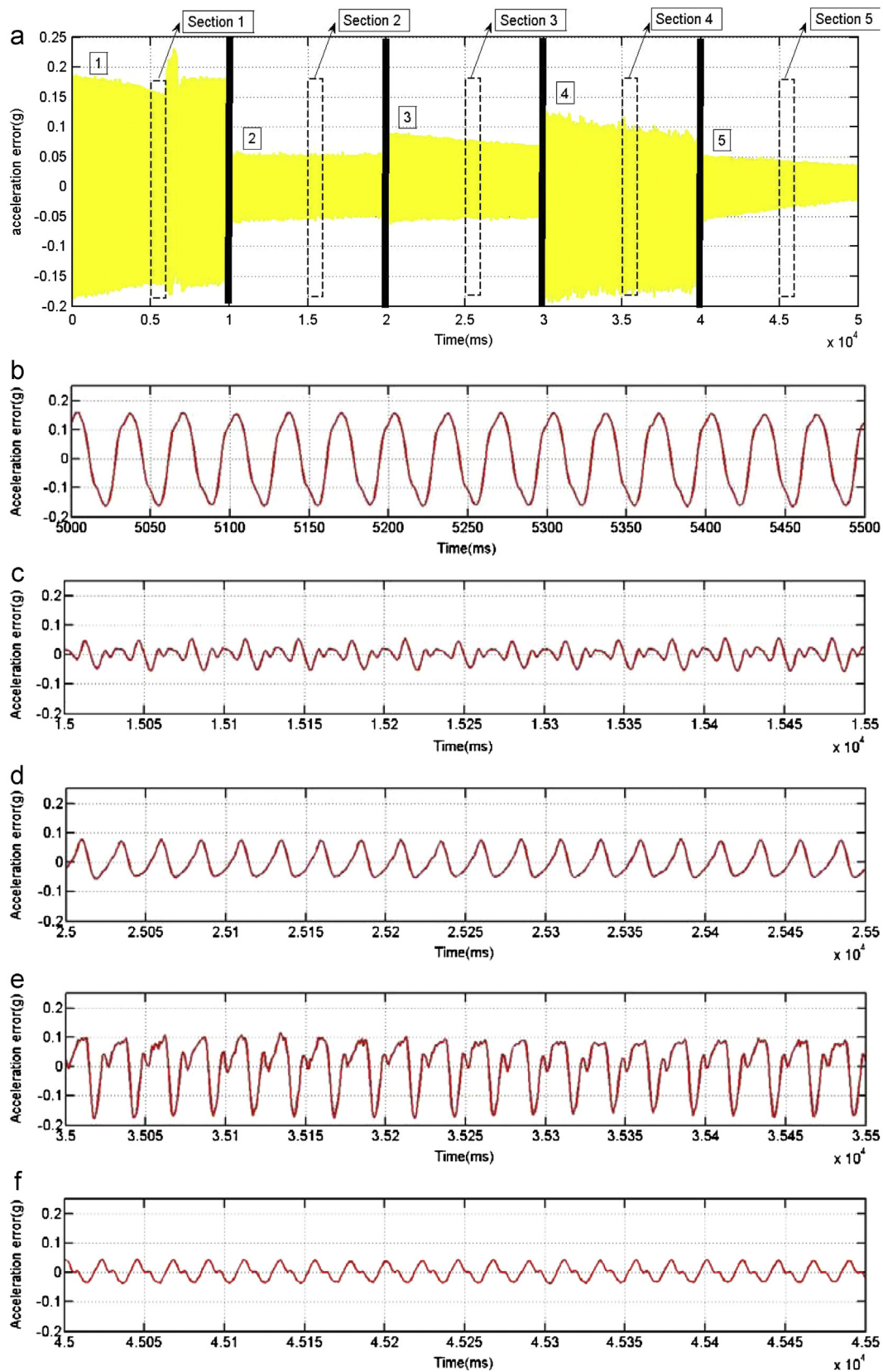
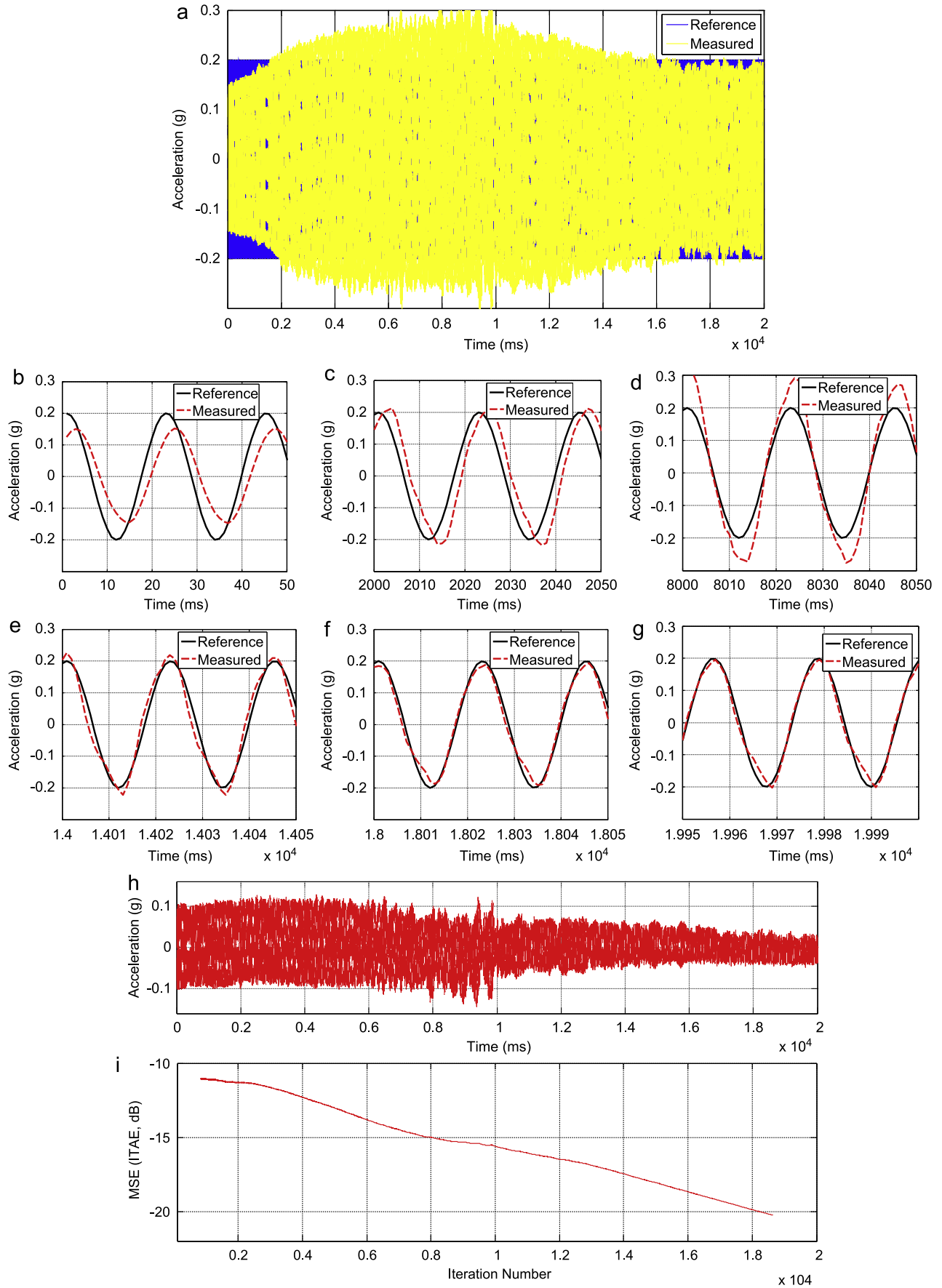


Fig. 15. Experimental results with different AICs according to different reference signal. (a) NLMS, (b) X-NLMS, (c) Err-NLMS, (d) proposed hybrid controller  $\beta=0$  and (e) proposed hybrid controller  $\beta=1$ .



**Fig. 16.** Acceleration tracking error with different AICs according to different reference signals. 1—NLMS (30 Hz), 2—X-NLMS (30 Hz), 3—Err-NLMS (40 Hz), 4—proposed controller (45 Hz, with  $\beta=0$ ), and 5—proposed controller (45 Hz, with  $\beta=1$ ). (a) Acceleration tracking error with different AICs, (b) Section 1, (c) Section 2, (d) Section 3, (e) Section 4 and (f) Section 5.



**Fig. 17.** Acceleration sine TWR with proposed hybrid controller. (a) TWR experimental with proposed controller, (b) 0–0.05 s, (c) 2–2.05 s, (d) 8–8.05 s, (e) 14–14.05 s, (f) 18–18.05 s, (g) 19.95–20 s, (h) tracking error and (i) ITAE experimental results.

The approximation to  $z^3$  based on Taylor series is employed in the test to improve the phase error caused by  $z^3$ . The compensator  $G_c(z)$  can be represented by the following equation:

$$z^3 = \frac{3.375z^4 - 8.138z^3 + 7.737z^2 - 3.407z + 0.5929}{z^4 - 1.472z^3 + 0.812z^2 - 0.1991z + 0.01832} \quad (17)$$

The ITF  $\hat{G}_a^{-1}(z)$  of the acceleration closed-loop system can be obtained as following:

$$\begin{aligned} \hat{G}_a^{-1}(z) = & -0.4 \left( \frac{z^2 - 1.726z + 0.7889}{z^2 - 1.609z + 0.7607} \right) \times \left( \frac{z^2 - 1.763z + 0.9916}{z^2 - 1.257z + 0.8616} \right) \\ & \times \left( \frac{z^2 + 1.149z + 0.7745}{z^2 - 0.4273z + 0.5441} \right) \left( \frac{z - 0.5679}{z} \right) \\ & \times \left( \frac{z^2 - 1.198z + 0.9467}{z^2} \right) \times \left( \frac{-5.032 + z^{-1}}{1} \right) \\ & \times \left( \frac{3.14 + 0.6574z^{-1} + z^{-2}}{1} \right) \\ & \times \left( \frac{3.375z^4 - 8.138z^3 + 7.737z^2 - 3.407z + 0.5929}{z^4 - 1.472z^3 + 0.812z^2 - 0.1991z + 0.01832} \right) \quad (18) \end{aligned}$$

A comparison between the magnitude frequency characteristics of  $\hat{G}_a(z)$  and  $G_a(z)$ , is presented in Fig. 12 for evaluating the

compensation performance. It can be seen from the figure that estimated inverse model can effectively compensate the acceleration closed-loop system dynamics in the interesting frequency range.

The transfer function of the acceleration modeling error  $\Delta G_a$  is also identified to compensate the modeling error using the proposed modeling error compensator. The comparison of the acceleration transfer function magnitude frequency characteristics between the experimental and estimated modeling error is presented in Fig. 13. It can be observed that the estimated modeling error is well approximated. However, the error between the model and actual system is still large. Therefore, a gain  $\beta$  is used to minimize the error and further to improve the effectiveness of the proposed modeling error compensator. Fig. 14 shows the ideal acceleration TWR errors with different values of  $\beta$  and it can be seen that an optimal TWR error can be obtained with  $\beta=1$ .

#### 4.3. Experimental results of the proposed hybrid controller

The TWR performance of the proposed hybrid controller is demonstrated in the third phase tests. The acceleration TWR

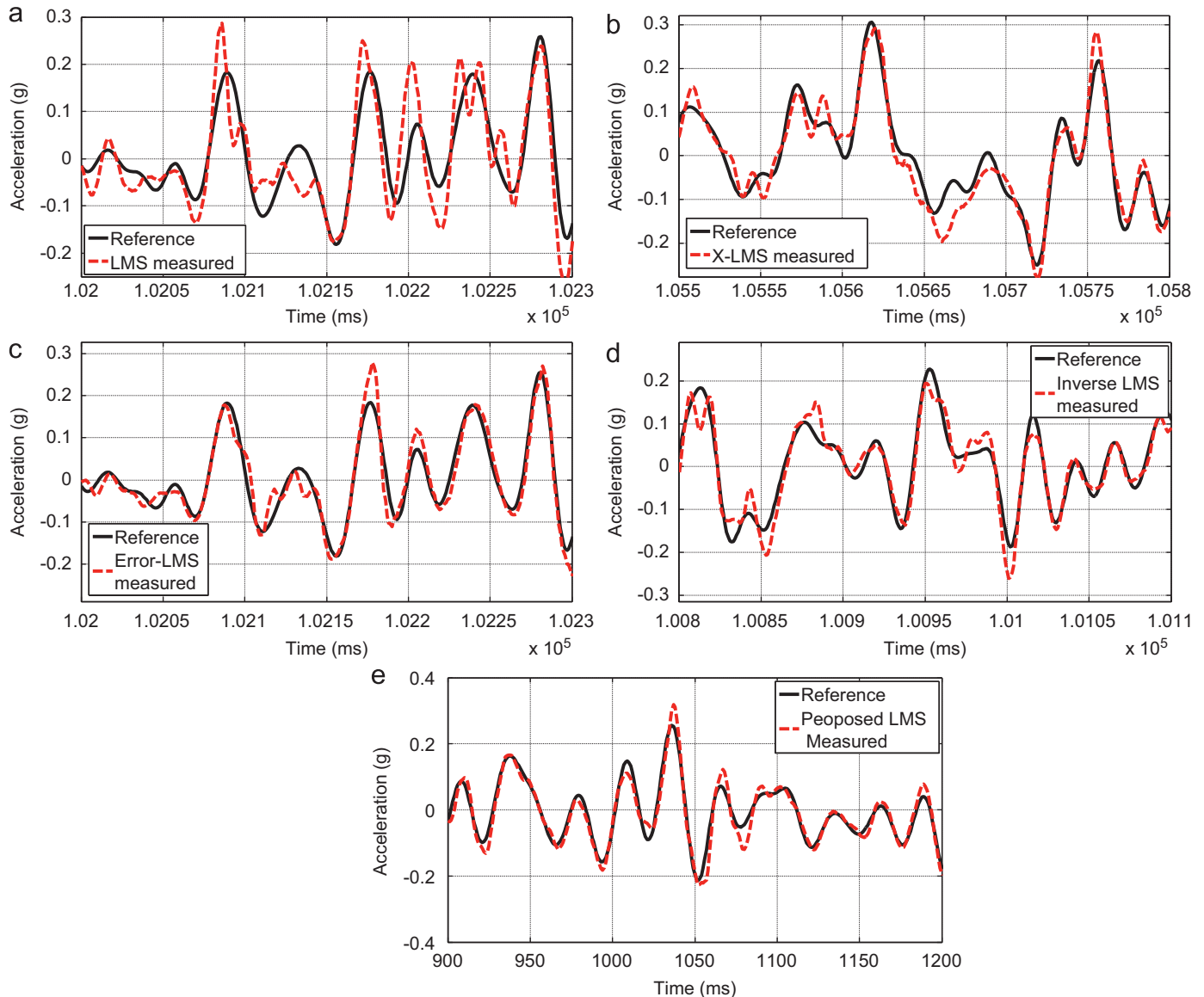


Fig. 18. Experimental results with different AICs according to 2–40 Hz random signal. (a) NLMS, (b) X-NLMS, (c) Err-NLMS, (d) Inv-NLMS and (e) proposed hybrid controller.

experimental results based on five adaptive algorithms (Du, Zhan, & Zhang, 2006; Lin & Chen, 2011): (a) the Normalized Least Mean Square (NLMS), (b) the Filtered-X NLMS, (c) the  $\varepsilon$ -Filtered NLMS, (d) the proposed hybrid controller with  $\beta=0$ , and (e) the proposed hybrid controller with  $\beta=1$  are compared. The sine signal with different frequencies are employed as the excitation signals of the different algorithms, named that: NLMS (30 Hz), X-NLMS (30 Hz), Err-NLMS (40 Hz) and proposed controller (45 Hz, with  $\beta=0$  and  $\beta=1$ ). Fig. 15 illustrates, on an enlarged time scale, the experimental results with different algorithms and different reference signals. It can be observed from the figure that a better TWR experimental result is achieved with the proposed hybrid controller.

Fig. 16 is the acceleration TWR error in Fig. 15. The convergence rate and TWR accuracy with different algorithms can be obtained from the figure. The acceleration TWR error reduces from 0.18g with conventional NLMS algorithm using 30 Hz sine signal to 0.03g with the proposed hybrid controller using 45 Hz sine signal. Therefore, these experimental results demonstrate the potential of the proposed hybrid controller to achieve better TWR accuracy.

In order to evaluate the TWR performance with the proposed hybrid control, Fig. 17 shows the comparison result between the acceleration command and measured acceleration output signal, including TWR and its partial enlarged drawing, tacking error, and integral of time-weighted absolute value of the error. The amplitude of the measured acceleration increases to around  $\pm 0.3g$  at 8th second and then decrease to around  $\pm 0.2g$  at 14th second. Close-up views of the comparison in Fig. 17(b–g) indicate that the acceleration output response converges to the acceleration command signal. The convergence performance can be observed from Fig. 17(h and i). The experimental results using a 45 Hz sine reference signal presented in Fig. 17 indicate that the accuracy of the TWR is improved.

A final set of experimental tests are carried out to evaluate the TWR accuracy using the proposed hybrid controller. The

acceleration command signal is a filtered random signal with a 2–40 Hz bandwidth. Fig. 18 illustrates, on an enlarged time scale, the experimental results with different algorithms with respect to 2–40 Hz bandwidth acceleration reference signal. Tests results revealed that a better TWR experimental result is achieved with the proposed hybrid controller. Fig. 19 is the acceleration frequency characteristics using different algorithms in Fig. 18. It can be clearly seen that the maximum magnitude frequency value reduces from  $\pm 7$  dB using NLMS to  $\pm 1.5$  dB using proposed hybrid controller after convergence, and that the maximum lag value increases from  $-40^\circ$  using the NLMS up to  $-7^\circ$  using the proposed hybrid controller after convergence to optimal solution. This is also reflected in Fig. 18, where the acceleration output response can be seen progressively to track the acceleration command more faithfully.

Consequently, these experiment results show that the proposed hybrid control strategy has yielded a more remarkable improvement in spite of the nonlinearity of EHST system, compared with the other controllers. So it can be concluded that the proposed hybrid control strategy is effective in the acceleration TWR and meets the experimental requirement of the EHST system.

## 5. Conclusions

In this study, an EHST was conducted to verify the tacking performance of the acceleration TWR. The proposed hybrid controller not only can extend the frequency bandwidth of an acceleration closed-loop system using the feed-forward compensation approach but also can improve the TWR accuracy using an online adaptive controller. Thus, the proposed hybrid controller is designed to combine the merits of feed-forward compensation and the AIC, and it is successfully applied in the EHST acceleration closed-loop system. To show the effectiveness of the proposed hybrid control strategy, some real experiments were conducted and the results proved the high TWR fidelity of the EHST system. Comparison studies with different control strategies have shown that the proposed hybrid control strategy has a better convergence rate and a higher TWR accuracy than the other controllers alone. These experimental results indicate that the proposed control strategy can successfully meet TWR requirements of the EHST.

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## Appendix A. Recursive extended least square algorithm

Here we shall be concerned with systems described by a Controlled Auto-Regressive Moving Average model (Cowan & Grant, 1985)

$$A(z^{-1})y(k) = B(z^{-1})u(k-d) + C(z^{-1})\xi(k) \quad (A1)$$

where  $\{y(k)\}$  and  $\{u(k)\}$  are the system input and output sequences, respectively,  $\{\xi(k)\}$  is a random noise sequence with a zero mean and an unknown variance, and  $A(z^{-1})$  and  $B(z^{-1})$  are the

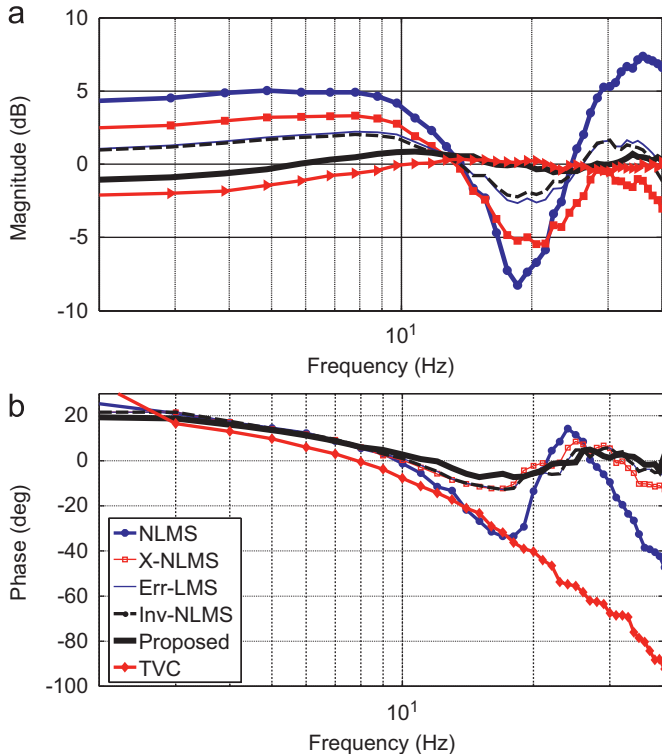


Fig. 19. Frequency property analysis of Fig. 18. (a) Magnitude property and (b) phase property.

polynomials of the backward shift operator  $z^{-1}$ ,  $[z^{-1}y(t)=y(t-1)]$ . These polynomials can be described as the following three equations:

$$A(z^{-1}) = 1 + a_1 z^{-1} + \dots + a_{n_a} z^{-n_a} \quad (A2)$$

$$B(z^{-1}) = b_0 + b_1 z^{-1} + \dots + b_{n_b} z^{-n_b} \quad (A3)$$

$$C(z^{-1}) = 1 + c_1 z^{-1} + \dots + c_{n_c} z^{-n_c} \quad (A4)$$

Assume that the system input–output data  $\{y(k)\}$  and  $\{u(k)\}$  are available, and the noise sequence  $\{\xi(k)\}$  is immeasurable, and the orders  $n_a$ ,  $n_b$  and  $n_c$  are known.

Write Eq. (A1) into the form of Least Square

$$y(k) = \varphi^T(k)\theta + \xi(k) \quad (A5)$$

where

$$\varphi(k) = [-y(k-1), \dots, -y(k-n_a), u(k), \dots, u(k-n_b), \xi(k-1), \dots, \xi(k-n_c)]^T,$$

$$\theta = [a_1, \dots, a_{n_a}, b_0, \dots, b_{n_b}, c_1, \dots, c_{n_c}]^T,$$

Since  $\xi(k)$  in  $\varphi(k)$  cannot be measured, its estimator  $\hat{\xi}(k)$  replaces it, presented as

$$\hat{\xi}(k) = y(k) - \hat{y}(k) = y(k) - \hat{\varphi}^T(k)\hat{\theta}(k) \quad (A6)$$

where

$$\hat{\varphi}(k) = [-y(k-1), \dots, y(k-n_a), \dots, u(k-d), \dots, u(k-d-n_b), \hat{\xi}(k-1), \dots, \hat{\xi}(k-n_c)]^T,$$

$$\hat{\theta} = [\hat{a}_1, \dots, \hat{a}_{n_a}, \hat{b}_0, \dots, \hat{b}_{n_b}, \hat{c}_1, \dots, \hat{c}_{n_c}]^T.$$

Replace  $\varphi(k)$  with  $\hat{\varphi}(k)$ , and the Recursive Extended Least Square equations with a forgetting factor can be expressed as (Cowan & Grant, 1985)

$$\hat{\theta}(k) = \hat{\theta}(k-1) + \mathbf{K}(k)[y(k) - \hat{\varphi}^T(k)\hat{\theta}(k-1)] \quad (A7)$$

$$\mathbf{K}(k) = \frac{\mathbf{P}(k-1)\hat{\varphi}(k)}{\lambda + \hat{\varphi}^T(k)\mathbf{P}(k-1)\hat{\varphi}(k)} \quad (A8)$$

$$\mathbf{P}(k) = [\lambda \mathbf{I} - \mathbf{K}(k)\hat{\varphi}^T(k)]\mathbf{P}(k-1) \quad (A9)$$

where  $k$  is the current time index,  $\lambda$  denotes the reciprocal of the exponential weighting factor,  $\mathbf{P}(k)$  is the inverse correlation matrix at step  $k$  and  $\mathbf{K}(k)$  is the gain vector at step  $k$ .

## Appendix B. Design of the inverse model

The identified EHST system acceleration model with the Recursive Extended Least Square algorithm may be a Zero Minimum Phase system caused by sensors and actuators (Rigney, Lucy, & Lawrence, 2009), and its inverse model is an unstable system. The accuracy of the TWR will be affected if the identified inverse model is a plant-model mismatch (Liu & He, 2007). So, how to improve the identification accuracy is critical.

Represent the acceleration closed-loop system with the following model based on Eq. (A1):

$$\hat{G}_a(z) = \frac{z^{-d}\hat{B}(z)}{\hat{A}(z)} \quad (B1)$$

where  $d$  is the number of pure time step delays.

In order to accurately design the acceleration inverse model in Fig. 7, it is helpful to set the factor  $\hat{B}(z)$  into Minimum Phase and Non-Minimum Phase zero polynomials

$$\hat{G}_a(z) = \frac{z^{-d}\hat{B}_a(z)\hat{B}_u(z)}{\hat{A}(z)} \quad (B2)$$

where  $\hat{B}_a(z)$  is the numerator polynomial including all plant

Minimum Phase zeros, and  $\hat{B}_u(z)$  contains all plant Non-Minimum Phase zeros.

Supposing that  $H(z) = \hat{B}_u(z)\hat{B}_u(z^{-1})$  is true, the following conclusions can be proved (Liu & He, 2007):

$$\angle H(e^{-j\omega T}) = 0, \quad \forall \omega \in R \quad (B3)$$

$$|\angle H(e^{-j\omega T})|^2 = \text{Im}^2[\hat{B}_u(e^{-j\omega T})] + \text{Re}^2[\hat{B}_u(e^{-j\omega T})], \quad \forall \omega \in R \quad (B4)$$

We can then express the identified acceleration inverse model  $\hat{G}_a^{-1}(z)$  as

$$\hat{G}_a^{-1}(z) = \alpha \frac{\hat{A}(z)\hat{B}_u(z^{-1})}{z^L \hat{B}_a(z)} \quad (B5)$$

where  $z^L$  is the sample delays, which has been added into  $\hat{G}_a^{-1}(z)$  to maintain the causality and  $\alpha$  is the gain for tuning the magnitude of the designed  $\hat{G}_a^{-1}(z)$  which can be represented as  $\alpha = \hat{B}_u^2(1)$ .

Non-causal inverse can result from the use of non-causal algorithms for approximating the unstable inverse of  $\hat{B}_u(z)$ . However, the added  $z^L$  may introduce a phase error between the designed inverse model and identified model. So, Taylor series approximation to  $z^L$  was employed in the paper to improve the phase error caused by  $z^L$ . The compensator  $G_c(s)$  can be represented by the following equation:

$$G_c(s) = e^{Ls} \cong \frac{1 + L \times T \times s/1 + (L \times T \times s)^2/2 + (L \times T \times s)^3/6 + (L \times T \times s)^4/24}{((s/1000) + 1)^4} \quad (B6)$$

The causal acceleration inverse transfer function  $\hat{G}_a(z)$  is

$$G_a^{-1}(z) = \alpha \frac{\hat{A}(z)\hat{B}_u(z^{-1})}{z^L \hat{B}_a(z)} G_c(z) \quad (B7)$$

where  $G_c(z)$  is the z-transform of the compensator  $G_c(s)$ .

## Appendix C. Optimum solution of the hybrid control

The Least Mean Square algorithm as an online feed-forward compensator is applied because of its simplicity, efficiency, and easy implementation (Plett, 2002, 2003). In Fig. 7,  $x_a(k)$  is the input vector of the Finite Impulse Response filter and is defined as  $x_a(k) = [x_a(k-1), \dots, x_a(k-M)]^T$ ,  $w(k)$  is the adaptive weighted factor and is defined as  $w = [w_0, \dots, w_{M-1}]$ , where,  $M$  is the length of the adaptive filter, and  $r_a(k)$  is the output vector of the Finite Impulse Response filter which can be expressed as the following equation:

$$\mathbf{r}_a(k) = \mathbf{w} \cdot \mathbf{x}_a(k) \quad (C1)$$

The desired acceleration reference  $d_a(k)$  is compared with acceleration measurement signal  $y_a(k)$ , and the acceleration TWR error signal  $e_a(k)$  can be represented as  $e_a(k) = d_a(k) - y_a(k)$ .  $e_a(k)$  and  $x_a(k)$  are inputted into the adaptive Least Mean Square algorithm for updating the adaptive filter weight coefficients  $w(k)$ , so as to minimize the mean-square errors. The convergence rate and steady state error of the AIC can be remarkably improved because the overall controlled plant of the AIC becomes a unit gain nearly and also a zero phase shift across the entire operating frequency range.

The objective of an adaptive algorithm is to iteratively find an optimal control in order to minimize the expected squared error. Referring to Fig. 7, the cost function  $J(k)$  can be defined as

$$J(k) = E[e_a^2(k)] \quad (C2)$$

where  $E[\cdot]$  denotes the statistical expectation.

When the reference model  $M(z)$  is taken into considered, substituting Eq. (7) to Eq. (C2), the objective function Eq. (C2)

yields

$$J(k) = E[(z^{-L} - w(k) - \hat{G}_a^{-1}(z)\Delta G_a w(k) + \beta \hat{G}_a^{-1}(z)\Delta G_a w(k) + \beta(\hat{G}_a^{-1}(z)\Delta G_a)^2 w(k))^2 x_a(k) x_a^T(k)] \quad (C3)$$

where the reference model  $M(z)$  is chosen as the response delay of the plant, which appropriately requires  $z^{-L}$  to be of the order of half the length of the adaptive filter (Aarón, 2005).

Let  $\mathbf{R}$  be an autocorrelation matrix defined as  $\mathbf{R} = E[x_a(k)x_a^T(k)]$ . The mean square error is nonnegative, so that  $\mathbf{R}$  is positive definite or positive semi-definite, and  $J(k)$  is a convex function of  $w(k)$ . As a result, its minimum value  $J_{\min}$  can be acquired by equating to zero its gradient with respect to  $w(k)$ , and gives (Widrow, 1986)

$$\begin{aligned} \frac{\partial J(k)}{\partial w(k)} &= -2E[(1 + \hat{G}_a^{-1}(z)\Delta G_a - \beta \hat{G}_a^{-1}(z)\Delta G_a - \beta(\hat{G}_a^{-1}(z)\Delta G_a)^2)z^{-L}]\mathbf{R} \\ &\quad + 2E[(1 + \hat{G}_a^{-1}(z)\Delta G_a - \beta \hat{G}_a^{-1}(z)\Delta G_a - \beta(\hat{G}_a^{-1}(z)\Delta G_a)^4)w(k)]\mathbf{R} \\ &= 0 \end{aligned} \quad (C4)$$

The optimum solution of the proposed control scheme is given by

$$w(k)_{\text{opt}} = (1 + \hat{G}_a^{-1}(z)\Delta G_a - \beta \hat{G}_a^{-1}(z)\Delta G_a - \beta(\hat{G}_a^{-1}(z)\Delta G_a)^2)^{-1} z^{-L} \quad (C5)$$

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